



DELHI PUBLIC SCHOOL NEWTOWN
SESSION: 2022-23
MONDAY TEST

CLASS : X
SUBJECT : CHEMISTRY

FULL MARKS : 40
DATE:15.6.2022

General Instructions:

- The paper consists of three printed pages.
- Answers should be to the point.
- Question numbers should be copied carefully while answering the questions.
- Marks will be deducted for spelling errors.

Question:1

Choose the correct answer from the options given below:

[5]

- (i) The metal which does not react with concentrated sodium hydroxide to produce hydrogen gas is:
A. Copper
B. Aluminium
C. Zinc
D. Lead
- (ii) Electron affinity is least for the element in:
A. Group 1
B. Group 2
C. Group 17
D. Group 18
- (iii) Which of the following is not a polar covalent molecule?
A. Ammonia
B. Water
C. Methane
D. Hydrogen chloride
- (iv) Which of the following pair of compounds will have the same empirical formula?
A. C_2H_2 and C_6H_6
B. $C_6H_{12}O_6$ and CH_3COCH_3
C. CH_4 and C_2H_6
D. CH_3CHO and C_2H_5OH
- (v) The metal whose hydroxide is soluble in both excess of sodium hydroxide and ammonium hydroxide is:
A. Lead
B. Zinc
C. Magnesium
D. Copper

Question:2

Name the following: [5]

- (i) The element in the third period whose electron affinity is zero.
- (ii) The element in the second period with smallest atomic radius.
- (iii) The metal whose hydroxide is soluble in excess of ammonium hydroxide but not in excess of sodium hydroxide.
- (iv) A covalent compound which undergoes ionisation when dissolves in water.
- (v) The element with highest electron affinity.

Question:3

Give reasons for the following: [5]

- (i) Alkali metals are strong reducing agents.
- (ii) Noble gases have very high ionisation potential.
- (iii) Carbon tetrachloride is a non conductor of electricity.
- (iv) Sodium chloride is a non conductor of electricity in solid state.
- (v) Lead hydroxide dissolves in excess sodium hydroxide solution.

Question:4

A hydrocarbon on analysis was found to contain 2.2% hydrogen, 26.6% carbon and 71.2% oxygen. If the relative molecular mass of the compound is 90, determine its molecular formula. [Atomic mass: C = 12, H = 1, O = 16] [5]

Question: 5

Given below is a section of the Periodic Table. The letters shown in the table are not the actual symbols of the elements. [5]

Group number	IA 1	IIA 2	IIIA 13	IVA 14	VA 15	VIA 16	VIIA 17	0 18
Period 1								F
Period 2						A	B	G
Period 3	D	J	E			C	I	
Period 4		H						

Answer the questions that follow:

- (i) The element with highest ionization potential.
- (ii) The element which is the most electronegative.
- (iii) The number of valence electrons present in E.
- (iv) Write the formula of the compound formed between E and B.
- (v) What type of bond is formed between D and I?

Question:6

- (a) Draw the electron dot and cross structure of the following: [3]

- (i) Formation Magnesium chloride
- (ii) Methane
- (iii) Hydronium ion

- (b) An element has 12 protons and 3 shells. [2]

- (i) How many valence electrons does this element have?

- (ii) To which group does it belong?

Question:7

- (a) Arrange the following as per the instructions given in the brackets () [3]
- (i) Chlorine, Nitrogen, Oxygen
(*increasing order of number of covalent bond*)
 - (ii) Na, Li, K
(*decreasing order of metallic character*)
 - (iii) Ar, He, Ne
(*increasing order of number of shells*)
- (b) Write a balanced chemical equation for each of the following: [2]
- (i) Ammonium hydroxide solution is added to aqueous ferrous sulphate solution.
 - (ii) Zinc granules are treated with concentrated potassium hydroxide solution.

Question:8

- (a) Differentiate between the following on the basis of the instruction given in the brackett: [3]
- (i) Halogens and Alkali metals (*position in the periodic table*)
 - (ii) Polar and non polar covalent bond (*charge separation*)
 - (iii) Ionisation and dissociation (*type of compounds*)
- (b) What would you observe when: [2]
- (i) Ammonium hydroxide solution is added to aqueous copper nitrate solution drop wise and then in excess.
 - (ii) Sodium hydroxide solution is added to aqueous zinc chloride solution in small quantities.